

Prelim Exam Activity

Marlon Alen I Flores
BSCS - 3A

CS311
2026-07-01

Code

```
1  using System;
2  using Avalonia.Controls;
3
4  namespace CS311_Prelim_Exam_Activity_Flores;
5
6  public partial class MainWindow : Window
7  {
8
9      Validators validators;
10     double grossPay = 0.0;
11     double tax = 0.0;
12     double sss = 0.0;
13     double pagibig = 0.0;
14     double philhealth = 0.0;
15     double totalDeductions = 0.0;
16     double netPay = 0.0;
17     double additionalPay = 0.0;
18     double totalPay = 0.0;
19
20
21     public MainWindow()
22     {
23         InitializeComponent();
24         validators = new Validators();
25     }
26
27     private void SubmitButton_Click(object sender,
28         Avalonia.Interactivity.RoutedEventArgs e)
29     {
30         double rate = 0;
31         double hours = 0;
32         if (validators.validateTextBox(in RateTextBox) is string s)
33         {
34             DataValidationErrors.SetError(RateTextBox, new CleanException(s));
35         }
36         else if (Double.TryParse(RateTextBox.Text, out rate) && (rate < 755 ||
37             rate > 3000))
38         {
39             DataValidationErrors.SetError(RateTextBox, new CleanException("Rate
40                 must be between 755 to 3000"));
41         }
42     }
43 }
```

```

38     }
39     else
40     {
41         DataValidationErrors.ClearErrors(RateTextBox);
42     }
43
44     if (validators.validateTextBox(in HoursTextBox) is string s2)
45     {
46         DataValidationErrors.SetError(HoursTextBox, new
47             CleanException(s2));
48     }
49     else if (Double.TryParse(HoursTextBox.Text, out hours) && (hours < 40
50         || hours > 200))
51     {
52         DataValidationErrors.SetError(HoursTextBox, new
53             CleanException("Hours must be between 40 to 200"));
54     }
55     else
56     {
57         DataValidationErrors.ClearErrors(HoursTextBox);
58     }
59
60     if (validators.countErrors() == 0)
61     {
62         grossPay = rate * hours;
63         tax = grossPay * 0.12;
64
65         sss = cbSSS.IsChecked.GetValueOrDefault(false) ? grossPay * 0.10 :
66             0.0;
67         pagibig = cbPagibig.IsChecked.GetValueOrDefault(false) ? grossPay *
68             0.08 : 0.0;
69         philhealth = cbPhilhealth.IsChecked.GetValueOrDefault(false) ?
70             grossPay * 0.06 : 0.0;
71
72         totalDeductions = sss + pagibig + philhealth;
73         netPay = grossPay - totalDeductions;
74
75         additionalPay = (bool)rbPermanent.IsChecked! ? 500.0 : 0.0;
76
77         totalPay = netPay + additionalPay;
78
79         GrossPayTextBox.Text = $"{grossPay:F2}";
80         TaxTextBox.Text = $"{tax:F2}";
81         SSSTextBox.Text = sss == 0.0 ? "" : $"{sss:F2}";
82         PagibigTextBox.Text = pagibig == 0.0 ? "" : $"{pagibig:F2}";
83         PhilhealthTextBox.Text = philhealth == 0.0 ? "" :
84             $"{philhealth:F2}";

```

```

79         TotalDeductionsTextBox.Text = $"{totalDeductions:F2}";
80         NetPayTextBox.Text = $"{netPay:F2}";
81         AdditionalPayTextBox.Text = $"{additionalPay:F2}";
82         TotalPayTextBox.Text = $"{totalPay:F2}";
83     }
84 }
85
86 private void ClearButton_Click(object sender,
87     Avalonia.Interactivity.RoutedEventArgs e)
88 {
89     RateTextBox.Text = "";
90     HoursTextBox.Text = "";
91     cbPagibig.IsChecked = false;
92     cbPhilhealth.IsChecked = false;
93     cbSSS.IsChecked = false;
94     rbContractual.IsChecked = true;
95     rbPermanent.IsChecked = false;
96
97     GrossPayTextBox.Text = "";
98     TaxTextBox.Text = "";
99
100     SSSTextBox.Text = "";
101     PagibigTextBox.Text = "";
102     PhilhealthTextBox.Text = "";
103
104     TotalDeductionsTextBox.Text = "";
105     NetPayTextBox.Text = "";
106     AdditionalPayTextBox.Text = "";
107     TotalPayTextBox.Text = "";
108
109     grossPay = 0.0;
110     tax = 0.0;
111     sss = 0.0;
112     pagibig = 0.0;
113     philhealth = 0.0;
114     totalDeductions = 0.0;
115     netPay = 0.0;
116     additionalPay = 0.0;
117     totalPay = 0.0;
118     DataValidationErrors.ClearErrors(RateTextBox);
119     DataValidationErrors.ClearErrors(HoursTextBox);
120 }
121 }

```

Validator

```
1  using System.Collections.Generic;
2  using System.Linq;
3  using Avalonia.Controls;
4  using Avalonia.Input;
5
6  namespace CS311_Prelim_Exam_Activity_Flores;
7
8  public class CleanException : System.Exception
9  {
10     public CleanException(string message) : base(message) { }
11
12     // Overriding this removes the "System.Exception:" prefix when printing the
13     // exception
14     public override string ToString()
15     {
16         return Message;
17     }
18 }
19
20 internal class Validators
21 {
22     private HashSet<TextBox> validatedTextBoxes = new HashSet<TextBox>();
23     private string errorMsg = "";
24
25     public string? validateTextBox(in TextBox textBox)
26     {
27         validatedTextBoxes.Add(textBox);
28         if (string.IsNullOrEmpty(textBox.Text))
29         {
30             errorMsg = $"{textBox.Name} is empty.";
31             return errorMsg;
32         }
33         else if (!double.TryParse(textBox.Text, out double value))
34         {
35             errorMsg = $"{textBox.Name} is not a valid number.";
36             return errorMsg;
37         }
38         else
39         {
40             return null;
41         }
42     }
43
44     public int countErrors()
45     {
46         int errorCount = 0;
```

```
46         foreach (var textBox in validatedTextBoxes)
47         {
48             errorCount += DataValidationErrors.GetErrors(textBox)?.Count() ?? 0;
49         }
50         return errorCount;
51     }
52
53     public void clearErrors()
54     {
55         foreach (var textBox in validatedTextBoxes)
56         {
57             DataValidationErrors.ClearErrors(textBox);
58         }
59         validatedTextBoxes.Clear();
60     }
61
62 }
```

Scenarios

1. All Inputs are empty, select permanent, and click submit.

Rate:		RateTextBox is empty.
Hours Worked:		HoursTextBox is empty.
Status:	☐ Contractual ☒ Permanent	
Deductions:	☐ SSS ☐ Pag-ibig ☐ Philhealth	
Submit		Clear
Gross Pay		
Tax		
SSS		
Pag-ibig		
Philhealth		
Total Deductions		
Net Pay		
Additional Pay		
Total Pay		

2. Rate is 20 and hours worked is empty, select permanent, and click submit.

Rate:	20
	Rate must be between 755 to 3000
Hours Worked:	
	HoursTextBox is empty.
Status:	☐ Contractual ☒ Permanent
Deductions:	☐ SSS ☐ Pag-ibig ☐ Philhealth
Submit Clear	
Gross Pay	
Tax	
SSS	
Pag-ibig	
Philhealth	
Total Deductions	
Net Pay	
Additional Pay	
Total Pay	

3. Rate is empty and hours worked is 20, select permanent, and click submit.

Rate:		RateTextBox is empty.
Hours Worked:	20	Hours must be between 40 to 200
Status:	☐ Contractual ☒ Permanent	
Deductions:	☐ SSS ☐ Pag-ibig ☐ Philhealth	
Submit		Clear
Gross Pay		
Tax		
SSS		
Pag-ibig		
Philhealth		
Total Deductions		
Net Pay		
Additional Pay		
Total Pay		

4. Rate and hours worked is 20, select permanent, and click submit.

Rate:	20
	Rate must be between 755 to 3000
Hours Worked:	20
	Hours must be between 40 to 200
Status:	☐ Contractual ☒ Permanent
Deductions:	☐ SSS ☐ Pag-ibig ☐ Philhealth
Submit Clear	
Gross Pay	
Tax	
SSS	
Pag-ibig	
Philhealth	
Total Deductions	
Net Pay	
Additional Pay	
Total Pay	

5. Rate is 755 and hours worked is 20, select permanent, and click submit.

Rate:	755
	Rate must be between 755 to 3000
Hours Worked:	20
	Hours must be between 40 to 200
Status:	☐ Contractual ☒ Permanent
Deductions:	☐ SSS ☐ Pag-ibig ☐ Philhealth
SubmitClear	
Gross Pay	
Tax	
SSS	
Pag-ibig	
Philhealth	
Total Deductions	
Net Pay	
Additional Pay	
Total Pay	

6. Rate is 20 and hours worked is 100, select permanent, and click submit.

Rate:	20
Rate must be between 755 to 3000	
Hours Worked:	100
Status:	☐ Contractual ☒ Permanent
Deductions:	☐ SSS ☐ Pag-ibig ☐ Philhealth
Submit Clear	
Gross Pay	
Tax	
SSS	
Pag-ibig	
Philhealth	
Total Deductions	
Net Pay	
Additional Pay	
Total Pay	

7. Rate is 755 and hours worked is 100, select contractual, and click submit.

Rate:	755
Hours Worked:	100
Status:	☒ Contractual ☐ Permanent
Deductions:	☐ SSS ☐ Pag-ibig ☐ Philhealth
Submit Clear	
Gross Pay	75500.00
Tax	9060.00
SSS	
Pag-ibig	
Philhealth	
Total Deductions	0.00
Net Pay	75500.00
Additional Pay	0.00
Total Pay	75500.00

8. Rate is 755 and hours worked is 100, select contractual, select SSS, and click submit.

Rate:	755
Hours Worked:	100
Status:	☒ Contractual ☐ Permanent
Deductions:	☐ SSS ☐ Pag-ibig ☐ Philhealth
Submit Clear	
Gross Pay	75500.00
Tax	9060.00
SSS	
Pag-ibig	
Philhealth	
Total Deductions	0.00
Net Pay	75500.00
Additional Pay	0.00
Total Pay	75500.00

9. Rate is 755 and hours worked is 100, select contractual, select SSS, select pagibig, and click submit.

Rate:	755
Hours Worked:	100
Status:	☒ Contractual ☐ Permanent
Deductions:	☒ SSS ☒ Pag-ibig ☐ Philhealth
Submit Clear	
Gross Pay	75500.00
Tax	9060.00
SSS	7550.00
Pag-ibig	
Philhealth	
Total Deductions	7550.00
Net Pay	67950.00
Additional Pay	0.00
Total Pay	67950.00

10. Rate is 755 and hours worked is 100, select contractual, select SSS, select, pagibig, select philhealth, and click submit.

Rate:	755
Hours Worked:	100
Status:	☒ Contractual ☐ Permanent
Deductions:	☒ SSS ☒ Pag-ibig ☒ Philhealth
Submit Clear	
Gross Pay	75500.00
Tax	9060.00
SSS	7550.00
Pag-ibig	
Philhealth	
Total Deductions	7550.00
Net Pay	67950.00
Additional Pay	0.00
Total Pay	67950.00

11. Rate is 755 and hours worked is 100, select permanent, and click submit.

Rate:	755
Hours Worked:	100
Status:	☐ Contractual ☒ Permanent
Deductions:	☐ SSS ☐ Pag-ibig ☐ Philhealth
Submit Clear	
Gross Pay	75500.00
Tax	9060.00
SSS	
Pag-ibig	
Philhealth	
Total Deductions	0.00
Net Pay	75500.00
Additional Pay	500.00
Total Pay	76000.00

12. Rate is 755 and hours worked is 100, select permanent, select SSS, and click submit.

Rate:	755
Hours Worked:	100
Status:	☐ Contractual ☒ Permanent
Deductions:	☒ SSS ☐ Pag-ibig ☐ Philhealth
Submit Clear	
Gross Pay	75500.00
Tax	9060.00
SSS	7550.00
Pag-ibig	
Philhealth	
Total Deductions	7550.00
Net Pay	67950.00
Additional Pay	500.00
Total Pay	68450.00

13. Rate is 755 and hours worked is 100, select permanent, select SSS, select pagibig, and click submit.

Rate:	755
Hours Worked:	100
Status:	☐ Contractual ☒ Permanent
Deductions:	☒ SSS ☒ Pag-ibig ☐ Philhealth
Submit Clear	
Gross Pay	75500.00
Tax	9060.00
SSS	7550.00
Pag-ibig	6040.00
Philhealth	
Total Deductions	13590.00
Net Pay	61910.00
Additional Pay	500.00
Total Pay	62410.00

14. Rate is 755 and hours worked is 100, select permanent, select SSS, select, pagibig, select philhealth, and click submit.

Rate:	755
Hours Worked:	100
Status:	☐ Contractual ☒ Permanent
Deductions:	☒ SSS ☒ Pag-ibig ☐ Philhealth
Submit Clear	
Gross Pay	75500.00
Tax	9060.00
SSS	7550.00
Pag-ibig	6040.00
Philhealth	
Total Deductions	13590.00
Net Pay	61910.00
Additional Pay	500.00
Total Pay	62410.00

Video

<https://drive.google.com/file/d/1DQiKDV6anC-tLUHXxwQsv4vTSNVI129w/view?usp=sharing>